Bootstrap Inference in the Presence of Bias

Cavaliere, Gonçalves, Nielsen & Zanelli

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Toulouse School of Economics — Econometrics Reading Group

Roadmap

1. **Introduction** — the bootstrap's third role
2. **Motivating examples** — model averaging, ridge, nonparametric regression
3. **General theory** — Theorem 3.1 and prepivoting
4. **Examples revisited** — implementing the fix
5. **Wrap-up**

Introduction

The bootstrap so far in this reading group

Two established roles:

Variance estimation When the asymptotic variance is unknown or hard to compute, the bootstrap provides a simulation-based approximation to the sampling distribution.

Asymptotic refinements When first-order asymptotics are available, the bootstrap can improve accuracy — reducing size distortion from $O(n^{-1/2})$ to $O(n^{-1})$ (Beran 1987, Hall 1992).

Use the bootstrap to **transform an invalid p-value into a valid one**, even when the estimator has a non-estimable asymptotic bias.

The general problem

Scalar parameter θ, estimator $\hat{\theta}_n$, normalised statistic:

$$T_n := g(n)(\hat{\theta}_n - \theta) \xrightarrow{d} B + \xi_1 \tag{1.1}$$

- $g(n) \to \infty$: convergence rate (e.g. $\sqrt{n}$, $(nh)^{1/2}$).
- ξ_1: centred noise with continuous cdf G.
- B: asymptotic bias — cannot be consistently estimated.

Bootstrap statistic $T_n^* := g(n)(\hat{\theta}_n^* - \hat{\theta}_n)$ satisfies:

$$T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1 \tag{1.2}$$

where $\hat{B}_n$ is the implicit bootstrap bias (not a deliberate estimator — it is what the bootstrap DGP produces automatically).

But the bootstrap bias does not match the true bias:

$$\hat{B}_n - B \xrightarrow{d} \xi_2 \neq 0 \tag{1.3}$$

Reading (1.1)–(1.3): the logical chain

(1.1): T_n has asymptotic bias B_n

↓

(1.2): bootstrap captures noise ξ_1 correctly

↓

(1.3): bootstrap bias $\hat{B}_n \neq B_n$ — gap is ξ_2

↓

$\Longrightarrow$ bootstrap p-value is **not uniform**

If instead $\hat{B}_n - B_n = o_p(1)$ (consistency), then $\xi_2 = 0$ a.s. and the bootstrap is valid in the standard sense. The paper addresses the case where this fails.

How bad is the distortion?

In all three examples, the bootstrap p-value satisfies $\hat{p}_n \xrightarrow{d} \Phi(m^{-1}\Phi^{-1}(U))$ for some $m \neq 1$.

If you reject at nominal level α, the actual rejection probability is $\Phi(m\,\Phi^{-1}(\alpha))$:

	$m = 1$	$m = 1.2$	$m = 0.8$	$m = 0.5$
Nominal $\alpha = 5\%$	5.0%	3.1%	7.8%	16.4%
Nominal $\alpha = 10\%$	10.0%	7.1%	13.6%	26.4%

A "5% test" can reject three times too often.

Monte Carlo evidence (Table B.1, supplement): standard bootstrap at nominal 5% rejects at $\approx 10\%$; the corrected $\tilde{p}_n$ is near-perfectly sized.

Motivating Examples

Common structure across examples

All three examples share the same pattern:

1. An estimator with asymptotic bias B_n that converges to a nonzero B.
2. The bootstrap generates an implicit bias $\hat{B}_n$ that differs from B_n by $\xi_2 \neq 0$.
3. The source: an inestimable quantity at the $n^{-1/2}$ boundary.

	Model Averaging	Ridge	Nonparametric
Rate $g(n)$	$n^{1/2}$	$n^{1/2}$	$(nh)^{1/2} \approx n^{2/5}$
Bias source	omitted variables	shrinkage to zero	smoothing curvature
Inestimable	δ (local to zero)	δ (local to zero)	$\beta''(x)$ (impractical)

Example 1: Model averaging

Model: $y = x\beta + Z\delta + \varepsilon, \;\; \varepsilon | W \sim \text{iid}(0, \sigma^2)$.

Fit M submodels with controls Z_m, average: $\tilde{\beta}_n = \sum_m \omega_m \tilde{\beta}_{m,n}$.

Using $\tilde{\beta}_n = \beta + Q_n\delta + A_n\varepsilon$:

$$T_n = n^{1/2}(\tilde{\beta}_n - \beta) = \underbrace{Q_n\, n^{1/2}\delta}_{B_n} + \underbrace{\bar{d}'_{M,n}\, n^{1/2} S_{W\varepsilon}}_{\xi_{1,n} \xrightarrow{d} N(0, v^2)}$$

Local-to-zero: $\delta = cn^{-1/2} \;\Rightarrow\; B_n \to Qc = B \neq 0$.

Why local-to-zero? If δ is fixed $\Rightarrow B_n$ diverges (too pessimistic). If $\delta = 0 \Rightarrow$ no bias (too optimistic). The rate $n^{-1/2}$ is the boundary where bias and noise are the same order — the interesting regime for model averaging.

$Q_n := \sum_m \omega_m S^{-1}_{xx.Z_m} S_{xZ.Z_m}; \quad \bar{d}'_{M,n} := \sum_m \omega_m S^{-1}_{xx.Z_m}(1, -S_{xZ_m} S^{-1}_{Z_m Z_m} R'_m)$.

Example 1: Why the bootstrap fails

FRB: generate $y^* = x\hat{\beta}_n + Z\hat{\delta}_n + \varepsilon^*$, compute $T_n^* = \hat{B}_n + n^{1/2}A_n\varepsilon^*$.

(1.2) holds: $T_n^* - \hat{B}_n = n^{1/2}A_n\varepsilon^* \xrightarrow{d^*}_p N(0, v^2)$. ✓

(1.3) — the gap:

$$\hat{B}_n - B_n = Q_n\, n^{1/2}(\hat{\delta}_n - \delta) = \bar{b}'_{M,n}\, n^{1/2} S_{W\varepsilon} \xrightarrow{d} \xi_2 \sim N(0, v_{22})$$

with $v_{22} > 0$ because $n^{1/2}(\hat{\delta}_n - \delta) = O_p(1)$, not $o_p(1)$.

Key insight: both $\xi_{1,n}$ and $\hat{B}_n - B_n$ are linear in the same score $n^{1/2}S_{W\varepsilon}$, just with different loadings $\bar{d}'_M$ vs. $\bar{b}'_M$. They are jointly normal and correlated.

Example 2: Ridge regression — setup

$y_t = \theta' x_t + \varepsilon_t, \;\; \varepsilon_t \sim \text{iid}(0, \sigma^2)$. Test $H_0 : g'\theta = r$.

Ridge estimator: $\tilde{\theta}_n = \tilde{S}_{xx}^{-1} S_{xy}, \;\; \tilde{S}_{xx} := S_{xx} + n^{-1} c_n I_p$.

Decomposition (using $\tilde{S}_{xx}^{-1} S_{xx} - I = -\frac{c_n}{n} \tilde{S}_{xx}^{-1}$):

$$T_n = n^{1/2}(g'\tilde{\theta}_n - r) = \underbrace{-\frac{c_n}{n}\, g' \tilde{S}_{xx}^{-1}\, \delta}_{B_n} + \underbrace{g' \tilde{S}_{xx}^{-1} \frac{X'\varepsilon}{n^{1/2}}}_{\xi_{1,n} \xrightarrow{d} N(0, v^2)}$$

With $\theta = \delta n^{-1/2}$ and $c_n/n \to c_0 > 0$: $\;B_n \to B = -c_0\, g' \tilde{\Sigma}_{xx}^{-1} \delta \neq 0$.

Variance: $v^2 = \sigma^2\, g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx} \tilde{\Sigma}_{xx}^{-1} g$.

The rates $\theta \sim n^{-1/2}$ and $c_n \sim n$ are calibrated so that $\frac{c_n}{n} \cdot n^{1/2}\theta = c_0 \delta = O(1)$ — bias and noise at the same order.

Example 2: Ridge — bootstrap failure

Pairs bootstrap: resample (y_t^*, x_t^*), compute $T_n^* = n^{1/2} g'(\tilde{\theta}_n^* - \hat{\theta}_n)$.

(Centre at OLS $\hat{\theta}_n$, not ridge $\tilde{\theta}_n$, to preserve $E^*[x_t^* \varepsilon_t^*] = 0$ — same principle as GMM bootstrap.)

(1.2) holds: $T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v^2)$ with $\hat{B}_n = -\frac{c_n}{n} g' \tilde{S}_{xx}^{-1} n^{1/2} \hat{\theta}_n$. ✓

(1.3) — the gap:

$$\hat{B}_n - B_n = -\frac{c_n}{n}\, g' \tilde{S}_{xx}^{-1}\, n^{1/2}(\hat{\theta}_n - \theta) \xrightarrow{d} \xi_2 \sim N(0, v_{22}), \quad v_{22} > 0$$

Again: $n^{1/2}(\hat{\theta}_n - \theta) = O_p(1)$, not $o_p(1)$.

Both statistics are linear in the same score $n^{1/2} S_{x\varepsilon}$:

$$\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} = \begin{pmatrix} g' \tilde{S}_{xx}^{-1} \\ -c_n n^{-1} g' \tilde{S}_{xx}^{-1} S_{xx}^{-1} \end{pmatrix} n^{1/2} S_{x\varepsilon} \xrightarrow{d} N(0, V)$$

Example 3: Nonparametric regression

$y_t = \beta(x_t) + \varepsilon_t$, fixed design $x_t = t/n$. Nadaraya–Watson with $h = cn^{-1/5}$:

$$T_n = (nh)^{1/2}(\hat{\beta}_h(x) - \beta(x)) = \underbrace{(nh)^{1/2}\left[\tfrac{1}{nh}\sum k_t\beta(x_t) - \beta(x)\right]}_{B_n} + \underbrace{(nh)^{-1/2}\sum k_t\varepsilon_t}_{\xi_{1,n}}$$

Taylor expansion: $B_n \to B = \frac{c^{5/2}}{2}\beta''(x)\kappa_2 \neq 0$.

Bootstrap: $y_t^* = \hat{\beta}_h(x_t) + \varepsilon_t^*$, centre at $\hat{\beta}_h(x)$.

$\hat{B}_n$ replaces $\beta(\cdot)$ by $\hat{\beta}_h(\cdot)$ — the bootstrap re-smooths an already-smoothed curve. The curvature $\hat{\beta}_h''(x)$ differs from $\beta''(x)$ by $O_p((nh)^{-1/2})$, which is of the same order as the bias itself at the MSE-optimal bandwidth. Hence $\hat{B}_n - B_n \xrightarrow{d} \xi_2 \neq 0$.

Unlike ridge and model averaging, $\beta''(x)$ *can* be estimated consistently — but doing so well enough to correct the bias requires a second bandwidth and performs poorly in finite samples (Calonico, Cattaneo, Farrell 2018).

General Results

Assumption 1 — the original statistic

Assumption (1)

$T_n - B_n \xrightarrow{d} \xi_1$, where ξ_1 is centred at zero with cdf

$$G(u) := \mathbb{P}(\xi_1 \leq u)$$

that is continuous and strictly increasing on its support.

Why these conditions on G?

- **Continuous:** needed for Polya's theorem (pointwise $\Rightarrow$ uniform convergence of cdfs).
- **Strictly increasing:** ensures G^{-1} is well-defined, which we need for the probability integral transform $G(\xi_1) \sim U[0,1]$.
- Not restricted to Gaussian — covers stable distributions, etc.

Assumption 2 — the bootstrap side

Assumption (2)

There exists D_n-measurable $\hat{B}_n$ such that:

(i) $T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1$

(ii) $\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}$, *with* $F(u) := \mathbb{P}(\xi_1 - \xi_2 \leq u)$ *continuous.*

- $\hat{B}_n$ is not a deliberate estimator of B_n — it is whatever bias the bootstrap DGP *implicitly generates*.
- Part (i): the bootstrap captures the noise correctly.
- Part (ii): joint convergence; ξ_2 quantifies the bias estimation failure.
- F must be continuous (but not necessarily strictly increasing) so that $H = F \circ G^{-1}$ is continuous.

Benchmark: what if $\hat{B}_n$ *were* consistent?

Suppose $\hat{B}_n - B_n = o_p(1)$, i.e. $\xi_2 = 0$ a.s. Then $F = G$ and:

$$\begin{aligned}
\hat{p}_n &= P^*(T_n^* \leq T_n) = P^*(T_n^* - \hat{B}_n \leq T_n - \hat{B}_n) \\
&= G(T_n - \hat{B}_n) + o_p(1) \qquad \text{(Assumption 2(i) + Polya)} \\
&\xrightarrow{d} G(\xi_1) \qquad \text{(since } T_n - \hat{B}_n = T_n - B_n + o_p(1) \xrightarrow{d} \xi_1\text{)} \\
&\sim U[0,1] \qquad \text{(probability integral transform, using } F = G\text{)}
\end{aligned}$$

$$T_n - \hat{B}_n \xrightarrow{d} \xi_1 - \xi_2 \sim F \neq G \implies \hat{p}_n \xrightarrow{d} G(F^{-1}(U)) \neq U.$$

Theorem 3.1 — the limit of $\hat{p}_n$

Theorem 3.1 (Cavaliere et al. 2024)

Under Assumptions 1 and 2,

$$\hat{p}_n \xrightarrow{d} G\big(F^{-1}(U_{[0,1]})\big).$$

Implications:

- $\hat{p}_n$ is **not uniform** unless $G = F$ (i.e. $\xi_2 = 0$).
- The cdf of $\hat{p}_n$ converges to $H(u) = F(G^{-1}(u))$, which is continuous (because G^{-1} continuous + F continuous).
- H does not depend on B — the unknown bias cancels. This is what makes the correction feasible.

Theorem 3.1 — proof

Step 1. Rewrite, subtracting $\hat{B}_n$ (constant under P^*):

$$\hat{p}_n = P^*(T_n^* - \hat{B}_n \leq T_n - \hat{B}_n)$$

By Assumption 2(i), $P^*(T_n^* - \hat{B}_n \leq u) \to G(u)$ *uniformly* (Polya). Hence:

$$\hat{p}_n = G(T_n - \hat{B}_n) + o_p(1)$$

Step 2. Decompose $T_n - \hat{B}_n = (T_n - B_n) - (\hat{B}_n - B_n)$. By Assumption 2(ii) and the continuous mapping theorem:

$$T_n - \hat{B}_n \xrightarrow{d} \xi_1 - \xi_2$$

Step 3. By continuity of G and continuous mapping: $G(T_n - \hat{B}_n) \xrightarrow{d} G(\xi_1 - \xi_2)$.

Step 4. Since $\xi_1 - \xi_2$ has continuous cdf F, the probability integral transform gives $\xi_1 - \xi_2 \stackrel{d}{=} F^{-1}(U)$. Substitute:

$$\hat{p}_n \xrightarrow{d} G(F^{-1}(U_{[0,1]})) \qquad \square$$

Three remarks

Some bootstrap schemes give $\hat{B}_n \xrightarrow{p} 0$, so $\xi_2 = -B$ (a constant). Then $\hat{p}_n \xrightarrow{d} G(G^{-1}(U) + B)$ — the distortion depends on B.

This is **worse** than ξ_2 random: with random ξ_2, H is free of B and prepivoting works; with $\xi_2 = -B$, it doesn't.

Lesson: prefer bootstrap schemes that *attempt* to capture the bias (even noisily) over those that structurally ignore it.

- **Remark 3.2.** Confidence intervals are similarly invalid: $P(\theta_0 \in CI_n^{1-\alpha}) = P(\hat{p}_n \leq 1 - \alpha) \neq 1 - \alpha$.
- **Remark 3.3.** When ξ_2 is nondegenerate, the bootstrap conditional distribution $\hat{L}_n$ is *random in the limit*: $\hat{L}_n(u) \to G(u - B - \xi_2)$.

Corollary 3.1 — prepivoting restores validity

Theorem 3.1 showed $\hat{p}_n$ has asymptotic cdf $H(u) = F(G^{-1}(u))$, which is continuous.

Corollary 3.1

Under the conditions of Theorem 3.1, $H(\hat{p}_n) \xrightarrow{d} U_{[0,1]}$.

Proof: probability integral transform applied to $\hat{p}_n$ with continuous cdf H. $\square$

The idea (Beran 1987, 1988):

- Apply H to $\hat{p}_n$ to "unwarp" the distorted distribution back to uniform.
- Beran used this for *refinements*; this paper uses it to rescue a broken bootstrap.

But H is unknown $\Longrightarrow$ need to estimate it.

Approach A: Plug-in

Suppose $H(u) = H_\gamma(u)$ depends on finite-dimensional γ, and $\hat{\gamma}_n \xrightarrow{p} \gamma$.

Corollary 3.2

If $H_\gamma(u)$ is continuous in (γ, u) and $\hat{\gamma}_n \xrightarrow{p} \gamma$, then

$$\tilde{p}_n := H_{\hat{\gamma}_n}(\hat{p}_n) \xrightarrow{d} U_{[0,1]}.$$

Proof: $|\tilde{p}_n - H(\hat{p}_n)| = |H_{\hat{\gamma}_n}(\hat{p}_n) - H_\gamma(\hat{p}_n)| \leq \sup_u |H_{\hat{\gamma}_n}(u) - H_\gamma(u)| \xrightarrow{p} 0$ by continuity in γ. Then Slutsky + Corollary 3.1. $\square$

Key point: we never estimate B. In the Gaussian case $H(u) = \Phi(m^{-1}\Phi^{-1}(u))$ with $m^2 = v_d^2/v^2$, a ratio of variances — no B appears.

Cheap: one standard bootstrap + one formula. But requires case-specific work to find H_γ.

Approach B: Double bootstrap

Estimate H nonparametrically by resampling *from the bootstrap data*.

Construction:

1. From each bootstrap sample D_n^*, draw a second-level sample D_n^{**}.
2. Compute T_n^{**} and the second-level p-value $\hat{p}_n^* := P^{**}(T_n^{**} \leq T_n^*)$.
3. Estimate H by $\hat{H}_n(u) := P^*(\hat{p}_n^* \leq u)$.
4. Modified p-value: $\tilde{p}_n := \hat{H}_n(\hat{p}_n) = P^*(\hat{p}_n^* \leq \hat{p}_n)$.

Assumption 3 ensures the double bootstrap replicates the distortion:

(i) $T_n^{**} - \hat{B}_n^* \xrightarrow{d^{**}}_{p^*} \xi_1$ (noise replicated at level 2)

(ii) $T_n^* - \hat{B}_n^* \xrightarrow{d^*}_{p} \xi_1 - \xi_2$ (bias gap replicated)

Theorem 3.2

Under Assumptions 1, 2, 3: $\tilde{p}_n = \hat{H}_n(\hat{p}_n) \xrightarrow{d} U_{[0,1]}$.

Corollary 3.3 — the practical shortcut

Corollary 3.3

Under Assumptions 1, 2, 3:

$$\tilde{p}_n = P^*(T_n^* - \hat{B}_n^* \ \leq \ T_n - \hat{B}_n) + o_p(1).$$

Translation: the double bootstrap modified p-value is asymptotically equivalent to a single-level bootstrap applied to the bias-corrected statistic $T_n - \hat{B}_n$.

Practical recipe:

1. Compute $\hat{B}_n$ from the original data (available in closed form for ridge, model averaging).
2. Compute $\hat{B}_n^*$ from each bootstrap sample.
3. Bootstrap p-value: $P^*(T_n^* - \hat{B}_n^* \leq T_n - \hat{B}_n)$.

No double bootstrap needed when $\hat{B}_n$, $\hat{B}_n^*$ are analytically available.

Examples Revisited

Model averaging — the fix

Gaussian limit: $(\xi_1, \xi_2)' \sim N(0, V)$ with V continuous in ω, σ^2, Σ_{WW} — all estimable.

$$H(u) = \Phi(m^{-1}\Phi^{-1}(u)), \quad m^2 = v_d^2/v^2$$

Plug-in: estimate $\hat{m}_n$ from sample analogues, then:

$$\tilde{p}_n = \Phi(\hat{m}_n^{-1}\,\Phi^{-1}(\hat{p}_n)) \xrightarrow{d} U_{[0,1]}$$

Double bootstrap: also valid (Lemmas 4.1–4.2 verify Assumption 3).

Pairs bootstrap: extra complication — $T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v^2 + \kappa^2)$ with $\kappa^2 > 0$ (bootstrap resamples regressors, adding variance). Requires the generalised Assumption 4 (Section 3.5), but the plug-in and double bootstrap still deliver valid $\tilde{p}_n$.

Local power against $\beta_0 = \bar{\beta} + an^{-1/2}$: same as the full-model OLS test — no power loss.

Ridge regression — the fix

Same Gaussian structure. The distortion parameter simplifies to:

$$m^2 = \frac{g'\Sigma_{xx}^{-1} g}{g'\tilde{\Sigma}_{xx}^{-1}\Sigma_{xx}\tilde{\Sigma}_{xx}^{-1} g}$$

(Derived using $\Sigma_{xx} + 2c_0 I + c_0^2\Sigma_{xx}^{-1} = \tilde{\Sigma}_{xx}\Sigma_{xx}^{-1}\tilde{\Sigma}_{xx}$.)

Plug-in: replace population quantities by sample analogues:

$$\hat{m}_n^2 = \frac{g' S_{xx}^{-1} g}{g'\tilde{S}_{xx}^{-1} S_{xx}\tilde{S}_{xx}^{-1} g}, \qquad \tilde{p}_n = \Phi(\hat{m}_n^{-1}\,\Phi^{-1}(\hat{p}_n))$$

Double bootstrap: also valid (Lemma 4.5).

Valid *irrespective* of whether θ is local-to-zero or fixed. When $c_0 = 0$: $m = 1$, no correction needed (ridge = OLS).

Nonparametric regression — the fix

Remarkable feature: the distortion m depends only on the kernel K:

$$m^2 = 4 + \frac{\int\left(\int K(s-u)K(s)\,ds\right)^2 du - 4\int K(u)\int K(u-s)K(s)\,ds\,du}{\int K(u)^2\,du}$$

σ^2 cancels in the ratio $m^2 = v_d^2/v^2$; $\beta''(x)$ never enters.

Plug-in: compute m once from K (a number), then:

$$\tilde{p}_n = \Phi(m^{-1}\Phi^{-1}(\hat{p}_n))$$

No estimation needed. No second bandwidth.

Comparison with alternatives:

- *Undersmoothing* ($nh^5 \to 0$): removes bias, but kills power against $\beta_0(x) = \bar{\beta} + an^{-2/5}$.
- *Robust bias correction* (Calonico–Cattaneo–Farrell): estimates β'' with second bandwidth. This paper avoids that entirely.

Conclusion

Take-aways

1. The standard bootstrap is first-order invalid under non-estimable bias (model averaging, ridge, nonparametric regression, heavy tails, panel FE...).
2. The asymptotic distribution of $\hat{p}_n$ is non-uniform but free of B: $\hat{p}_n \xrightarrow{d} G(F^{-1}(U))$.
3. Apply **prepivoting** — transform $\hat{p}_n$ by H — to recover uniformity.
4. Two implementations: **plug-in** (when H_γ is tractable) and **double bootstrap** (automatic).
5. Shortcut (Corollary 3.3): bootstrap $T_n - \hat{B}_n$ directly when $\hat{B}_n$ is available in closed form.
6. No power loss; near-nominal size in finite samples (Table B.1).

The supplement contains two additional examples: a heavy-tailed location model (non-Gaussian ξ_1, unknown convergence rate) and dynamic panel data with incidental parameter bias (where the double bootstrap *fails* because $\hat{B}_n = 0$).

Thank you

Questions?

Cavaliere, Gonçalves, Nielsen & Zanelli (2024)
Bootstrap Inference in the Presence of Bias
JASA, 119(548), 2908–2918

Appendix: Gaussian special case

When $\xi_1 \sim N(0, v^2)$ and $(\xi_1, \xi_2) \sim N(0, V)$:

$$G(u) = \Phi(u/v), \quad F(u) = \Phi(u/v_d), \quad v_d^2 = v_{11} + v_{22} - 2v_{12}$$

$$H(u) = \Phi(m^{-1}\Phi^{-1}(u)), \quad m = v_d/v$$

Modified p-value: $\tilde{p}_n = \Phi(\hat{m}_n^{-1}\Phi^{-1}(\hat{p}_n))$.

Covers Examples 1–3 and Example 5 (panel). Example 4 (heavy tails) uses $G(u) = \Psi_\alpha(\omega^{-1}u)$ with Ψ_α the stable cdf.

Appendix: Section 3.5 — random centering (pairs bootstrap)

Some bootstrap schemes (e.g. pairs in model averaging) give $T_n^* - B_n^* \xrightarrow{d^*}_p \xi_1$ where B_n^* depends on bootstrap data.

Assumption 4 replaces 2(i): $T_n^* - \hat{B}_n \xrightarrow{d^*}_p \zeta_1$ with cdf $J \neq G$.

Theorem 3.4: $\hat{p}_n \xrightarrow{d} J(F^{-1}(U))$ and $H(u) = F(J^{-1}(u))$.

Plug-in and double bootstrap go through with J replacing G.

Caution: the plug-in formula for m is bootstrap-scheme-specific. If you derived m for FRB and applied it to pairs, you'd get the wrong correction. The double bootstrap handles this automatically.

Appendix: Power (Theorem 3.3)

Test $H_0 : \theta = \bar{\theta}$ vs. $H_1 : \theta < \bar{\theta}$.

Under local alternatives $\theta_0 = \bar{\theta} + a/g(n)$ with $a < 0$:

$$P(H(\hat{p}_n(\bar{\theta})) \leq \alpha) \;\rightarrow\; F(F^{-1}(\alpha) - a)$$

- Fixed alternatives ($a \to -\infty$): test is consistent.
- Local alternatives: power depends on F only, not G.
- Same power as an oracle test that knows B.

Appendix: Supplement examples

Example 4 — Heavy tails: $y_t = \theta + \varepsilon_t$ with $\varepsilon_t \sim S(\alpha)$, $\alpha \in (1, 2)$ (infinite variance).

- Non-Gaussian ξ_1, unknown convergence rate $n^{1-1/\alpha}$.
- Plug-in works: $H(u) = \Psi_\alpha(\omega \Psi_\alpha^{-1}(u))$.
- m-out-of-n bootstrap fails (zeros out bias $\Rightarrow$ Remark 3.1 applies).

Example 5 — Dynamic panel: Fixed effects MLE with incidental parameter bias.

- Cross-sectional pairs bootstrap gives $\hat{B}_n = 0 \;\Rightarrow\;$ distortion depends on B (Remark 3.1).
- But B *can* be estimated consistently here (Hahn–Kuersteiner 2011).
- Plug-in works; double bootstrap fails (Remark B.4: both levels zero out the bias).